measurements would get the same answers. To Newton, such facts about motion through space and time were as objective as two plus two equaling four. According to Newton, then, Bart will agree with Lisa and will report that the light beam was speeding away from him at 170 million miles per hour.

But when Bart returns, he doesn't agree at all. Instead, he dejectedly claims that no matter what he did—no matter how much he pushed the skateboard's limit—he saw the light speed away at 670 million miles per hour, not a bit less.[3] And if for some reason you don't trust Bart, bear in mind that thousands of meticulous experiments carried out during the last hundred years, which have measured the speed of light using moving sources and receivers, support his observations with precision.

How can this be?

Einstein figured it out, and the answer he found is a logical yet profound extension of our discussion so far. It must be that Bart's measurements of distances and durations, the input that he uses to figure out how fast the light is receding from him, are different from Lisa's measurements. Think about it. Since speed is nothing but distance divided by time, there is no other way for Bart to have found a different answer from Lisa's for how fast the light was outrunning him. So, Einstein concluded, Newton's ideas of absolute space and absolute time were wrong. Einstein realized that experimenters who are moving relative to each other, like Bart and Lisa, will not find identical values for measurements of distances and durations. The puzzling experimental data on the speed of light can be explained only if their perceptions of space and time are different.

Subtle but Not Malicious

The relativity of space and of time is a startling conclusion. I have known about it for more than twenty-five years, but even so, whenever I quietly sit and think it through, I am amazed. From the well-worn statement that the speed of light is constant, we conclude that *space and time are in the eye of the beholder.* Each of us carries our own clock, our own monitor of the passage of time. Each clock is equally precise, yet